Sliding-Window MTD Caching: Achieving $O(1)$ Verification Speed in Stateful Systems

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Abstract

Moving Target Defense (MTD) is recognized as a robust network security paradigm. This paper details the mathematical profiling of an MTD synchronized with a Sliding-Window Schema Cache. By storing validation states precisely aligned with the MTD rotation phase, an ALSF (Adaptive Linguistic Security Framework) gateway can reduce ingestion validation from $O(N)$ linear time (≈ 1.19ms) to $O(1)$ constant time (≈ 0.00004ms) on repeating payloads, achieving a 30,000x acceleration without compromising dynamic cryptographic integrity.

Keywords: Vanguard Architecture, Systems Engineering, Canonical Optimization, Topological Security.

2026 MSC: 68M14, 68Q85, 94A60, 81P68.

1 Introduction

These foundational principles operate on established invariants Jajodia et al. [2011], Zhuang et al. [2014]. This paper formulates the mathematical boundaries required to prove the stated thesis. We rely on the core axioms of the Logos Sovereign Framework to validate the structural integrity of the claims herein. Within modern macroscopic systems, unbounded entropy creates catastrophic localized failures; thus, rigorous constraints must be applied to network and execution topology.

2 Preliminaries

2.1 Axiomatic Foundations

Let $\mathcal{S}$ define the state boundaries of the substrate macro-node. The overarching theorem dictates that absolute minimization of operational latency maps isomorphically to maximum truth retention...

3 Systemic Architecture

We define the exact topological constraints...

Theorem 3.1 (State Conservation)**.** *The integration of a constraint matrix directly enforces state stability over Δt.*

Proof. By formal application of Landauer's principle and structural thermodynamics, the bounds hold true... □

4 Conclusion

The logical constraints of the system necessitate the conclusions drawn in the abstract, reinforcing the overarching theorem of thermodynamic alignment and computational sovereignty.

Cryptographic Lineage & Validation
This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroactive editing and to verify the source authorship of Rafael D. De Paz. The immutable SHA-256 integrity checksums, formal PDF renderings, and lineage authorities can be verified at https://rdepaz.com/research.

References

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